Regression Functional Forms: Exact Percent Change Formulas

1. Level-log Model

Equation: $Y = \beta_0 + \beta_1 \ln(X) + \epsilon$

Exact absolute change in Y, given X moves from X_1 to X_2:

$$\Delta Y = Y_2 - Y_1 = \beta_1 \times (\ln(X_2) - \ln(X_1))$$

Special case: 1% increase in X, i.e., $1.01 \times X_1$:

$$\Delta Y = \beta_1 \ln(1.01) \sim \beta_1 0.01 = \frac{\beta_1}{100}$$

2. Log-level Model

Equation: $\ln(Y) = \beta_0 + \beta_1 X + \epsilon$

Exact multiplicative given X moves from X_1 to X_2:

$$\ln\left(\frac{Y_2}{Y_1}\right) = \beta_1(X_2 - X_1)$$

$$Y_2/Y_1 = \exp(\beta_1(X_2 - X_1))$$

$$Y_2 = Y_1 \exp(\beta_1(X_2 - X_1))$$

Exact Percent Change in Y:

$$\Delta Y\% = (Y_2 - Y_1)/Y_1 \times 100\% = ((Y_1 \exp(\beta_1(X_2 - X_1) - Y_1))/Y_1) \times 100\% = (\exp(\beta_1 (X_2 - X_1)) - 1) \times 100\%$$

Special case: $X_2 - X_1 = 1$ and β_1 is small, say $\beta_1 = 0.1$

$$\Delta Y\% = (\exp(0.1) - 1) \times 100\% \sim 10\%$$

3. Log-log model

Equation: $\ln(Y) = \beta_0 + \beta_1 \ln(X) + \epsilon$

Exact multiplicative given X moves from X_1 to X_2:

$$\ln\left(\frac{Y_2}{Y_1}\right) = \beta_1 \ln\left(\frac{X_2}{X_1}\right)$$

$$\frac{Y_2}{Y_1} = \left(\frac{X_2}{X_1}\right)^{\beta_1}$$

$$Y_2 = Y_1 \left(\frac{X_2}{X_1}\right)^{\beta_1}$$

Exact percentage change:

$$\Delta Y\% = (Y_2 - Y_1)/Y_1 \times 100\% = Y_1 \left(\frac{X_2}{X_1}\right)^{\beta_1} - Y_1) \times 100\%$$

$$= (\left(\frac{X_2}{X_1}\right)^{\beta_1} - 1) \times 100\%$$

Special case: 1% increase in X, i.e., $1.01 \times X_1$ and small β_1:

$$\Delta Y\% = (1.01^{\beta_1} - 1) \times 100\% = (1.01\beta_1 - 1) \times 100\% = \beta_1 \times 1\%$$

(Because $1.01^{\beta_1} = 1 + 0.01\beta_1$ for small β_1)